# Notes: Humor as Entropy (Westbury et al., 2015)

## 1. Paper at a glance

Made-up words built from rarer letters (lower unigram entropy) are judged funnier, and this can be predicted before any human sees the words.

| | |
|---|---|
| Title | Telling the world's least funny jokes: On the quantification of humor as entropy |
| Authors | Chris Westbury, Cyrus Shaoul, Gail Moroschan, Michael Ramscar |
| Affiliations | University of Alberta (Psychology); University of Tübingen (Linguistics) |
| Venue | Journal of Memory and Language, 2015 |
| Field | Psycholinguistics, humor research, information theory |

**Where the idea came from:** while testing aphasic patients and running lexical decision tasks, the authors noticed some non-words reliably made people laugh.

**Research question:** Why are some non-words funny, and can we predict which ones?

**Theory tested:** Schopenhauer (1818). Humor = a violated conceptual expectation; the bigger the violation, the funnier.

**How they made it measurable:** a non-word violates the expectation "this will be a real word." How far it is from word-like can be measured with Shannon entropy over its letters.

**Three studies:**

1. **Study 1 (survey):** 968 people rated 5,928 non-words. People agree on which are funny.
2. **Experiment 1 (forced choice):** pick the funnier of two. Bigger entropy gap → more predictable choice.
3. **Experiment 2 (rating):** rate one word at a time. Lower entropy → higher rating ($r = -0.83$).

**Bonus check:** Dr. Seuss's invented words have lower entropy than ordinary English words.

**Claim to remember:** this is the first quantitative, information-theoretic test of Schopenhauer's 200-year-old theory.

## 2. Key terms

| Term | Simple meaning | Example |
|---|---|---|
| Non-word (NW) | A letter string that looks like a word but isn't one | *inquenter*, *subvick* |
| Pseudohomophone | A non-word that sounds like a real word (removed from stimuli) | *phish* (sounds like fish) |
| Bleached semantics | Words with very little meaning of their own | *and*, *if*, *but*; non-words even more so |
| Incongruity | Two things appearing together that don't normally go together | Chaplin's tramp: fine suit, clumsy behavior |
| Expectation violation | An event clashes with a concept you were applying to it | A pun where a word suddenly has a second meaning |
| n-gram | A chunk of n consecutive letters | In *cat*: unigrams c, a, t; bigrams ca, at; trigram cat |
| Unigram | A single letter | *e*, *q* |
| Markov chain (here) | Building a string piece by piece, where each next chunk must follow naturally from the previous one | Overlapping trigrams → *inquenter* |
| Surprisal | How surprising one event is; rarer = more surprising | Seeing *q* is more surprising than seeing *e* |
| Shannon entropy | Average surprise across a set of events | Fair coin = 1 bit; 99%-heads coin ≈ 0.08 bits |

| Term | Simple meaning | Example |
|---|---|---|
| Unigram entropy (of a string) | Sum of each letter's entropy contribution; low = built from rare letters | *subvick* ≈ 1.04 (low), *tessina* ≈ 2.02 (high) |
| Likert scale | A rating scale with fixed points | 1 = Not at all funny … 7 = Definitely funny |
| Forced choice | You must pick one of two options | Which is more humorous: *subvick* or *octeste*? |
| Monte Carlo analysis | Testing something by repeating random samples many times | 5,000 random pairs of raters |
| Mixed-effects model | Regression that separates the effect you care about from differences between people and items | Entropy effect, with participant as a random effect |
| AIC | Score for comparing models; lower = loses less information | AIC drop 3579 → 3511 = much better model |
| Cohen's d | Size of the difference between two groups | d ≈ 0.9 = large effect |
| Semantic parasitism | A non-word is funny because it borrows meaning from a real word | *shart*, *dongl* (resemble rude words) |

## 3. Background: theories of humor

The paper builds on incongruity theory, sharpened by Schopenhauer: humor needs an expectation that gets broken, not just something odd.

### Starting point: William James (1890)

Words carry feelings, not just meaning: "a feeling of *and*, a feeling of *but*." James chose function words because their meaning is weak, so feeling stands out. Non-words have even less meaning, so any feeling they evoke (like humor) should be easy to isolate.

## How the theory developed

| Thinker | Idea | Problem / example |
|---|---|---|
| Plato, Aristotle, Hobbes | Humor = laughing at others' flaws (derisive humor) | Many funny things involve no one's flaws |
| Hutcheson (1725) | Humor = incongruity between something dignified and something mean | Charlie Chaplin's well-dressed tramp |
| Bain (1865) | Objection: many incongruities aren't funny | A corpse at a feast, a decrepit man under a load, parental cruelty → pain or sadness, not laughter |
| Schopenhauer (1818) | Humor = sudden incongruity between a **concept** and the real objects thought through it | Fixes Bain: a fly in ointment is odd but breaks no concept, so not funny |
| Lipps (1898) | Non-words are funny because we expect words to mean something, and they don't | We're "tricked for one moment" |
| Hurley, Dennett & Adams (2011) | Humor rewards the brain for catching inconsistent beliefs ("cognitive debugging") | Evolutionary account of why violations feel good |
| Chafe (2007) | Humor = reacting to things it would be counterproductive to take seriously | Authors: belief-checking feels funny when the belief isn't worth keeping |

## Schopenhauer's two forms of funny incongruity

1. **Conceptual bifurcation** (authors' term): something you thought belonged to one category turns out to belong to two.
2. **Conceptual subsumption:** two different things turn out to fit under one concept.

**His two key claims:**

- Humor is about **patterned co-occurrence**: an event against an expectation, not rarity alone.
- **The bigger the incongruity, the funnier.** This is what the paper tests.

### Examples from the paper

- **Funny:** "When the clock is hungry it goes back four seconds." *Four/for* and *seconds* each have two meanings, so one clear expectation breaks.
- **Not funny, just confusing:** "When the clock is hungry, it rides a horse." Odd, but no single expectation to resolve.
- **Not getting a joke** feels bad because you sense a violation but can't find which expectation broke.
- **Corpse at a dinner party:** not funny. **Corpse tied up in a tie-dye shirt:** maybe funny, because two concepts are being forced together.

### Why use non-words?

In real jokes, probabilities can't be measured ("how often do a priest, a rabbi and a monk walk into a bar?"). With non-words, the relevant probabilities (letter frequencies) are exact.

## 4. Shannon entropy and letter entropy

A string's unigram entropy is the sum of each letter's contribution $-p \log_2 p$; rare letters contribute little, so strings of rare letters score low and are predicted funnier.

### The formulas

Surprisal of one event (in bits):

```
\text{surprisal}(x) = -\log_2 p(x)
```

Shannon entropy of a set of events:

```
H = -\sum_k p(x_k)\,\log_2 p(x_k)
```

### Coin example (from the paper)

- **Fair coin:** $-[0.5 \times \log_2 0.5 + 0.5 \times \log_2 0.5] =$ **1 bit**. You can't guess better than 50/50.
- **99%-heads coin:** $-[0.99 \times \log_2 0.99 + 0.01 \times \log_2 0.01] = 0.0143 + 0.0664 =$ **0.0807 bits**. Very predictable.
- **Key point:** the rare 1% outcome contributes only 0.066, while a 50% outcome contributes 0.5. **Rare components add little to total entropy.**

## Computing it for letters (3 steps)

1. **Probability:** count each letter in a big English corpus. p(letter) = count of that letter / total letters.
2. **Contribution:** for each letter, compute $-p \times \log_2 p$.
3. **Score the string:** add the contributions of every letter in it (repeated letters counted each time).

| Letter | p (approx.) | Surprisal $-\log_2 p$ | Contribution $-p \log_2 p$ |
|---|---|---|---|
| e | 0.127 | 2.98 | 0.378 |
| t | 0.091 | 3.46 | 0.315 |
| a | 0.082 | 3.61 | 0.296 |
| s | 0.063 | 3.99 | 0.251 |
| b | 0.015 | 6.06 | 0.091 |
| v | 0.0098 | 6.67 | 0.065 |
| k | 0.0077 | 7.02 | 0.054 |
| q | 0.001 | 9.97 | 0.010 |

Surprisal and contribution move in **opposite directions**: rare letters are very surprising but contribute little.

## Worked example (stimuli from the paper)

- **subvick** = 0.25 (s) + 0.14 (u) + 0.09 (b) + 0.07 (v) + 0.27 (i) + 0.14 (c) + 0.05 (k) ≈ **1.02** (paper: 1.04) → rare letters → **predicted funny**
- **tessina** = 0.32 (t) + 0.38 (e) + 0.25 (s) + 0.25 (s) + 0.27 (i) + 0.26 (n) + 0.30 (a) ≈ **2.02** (paper: 2.02) → common letters → **predicted unfunny**

**Memory rule:** low entropy = rare letters = less word-like = funnier.

## Small print

- A single letter doesn't strictly "have" an entropy; entropy belongs to the whole distribution. Each letter has a **contribution**. All 26 contributions sum to about 4.1–4.2 bits for English.

- The paper says entropy was divided by length, but the experiment values (1.04–2.09 for 7-letter words) match the **summed** version. The Dr. Seuss figure (~0.2–0.3) is **per letter**. All experiment words were 7 letters, so results are unaffected.

## 5. Study 1: Are non-words funny?

People reliably agree on which non-words are funny, even after removing strings that resemble rude words.

### Setup

- **Participants:** 968 University of Alberta undergraduates; online survey; partial course credit; ethics approved.
- **Design:** 60 groups, each rating ~100 non-words; same order within a group.
- **Scale:** 1 = Not at all funny … 7 = Definitely funny.
- Each non-word got ~16 ratings. Surveys not finished within 24 h were excluded.

### How the stimuli were made (LINGUA)

- **LINGUA** (Language-Independent Neighborhood Generator of the University of Alberta) builds non-words by Markov-chaining letter n-grams from a frequency dictionary.
- They used **trigrams**: every 3 consecutive letters appear in some real English word. Result: word-like, pronounceable strings, e.g. *inquenter*, *artorts*.
- **Dictionary:** 111,627 word types from a USENET corpus of ~7.8 billion tokens.
- 1,200 strings per length, 5 to 9 letters. Real words and pseudohomophones removed → **5,928 strings**.

### Result 1: ratings vary consistently

- Range: **1.1** (*exthe*, *suppect*) to **5.92** (*whong*). Extremes this far apart require agreement.
- Distribution follows a **power law**: 43.3% of all ratings were "1" (Fig. 2, r = .91).

### Result 2: Monte Carlo agreement test

A plain split-half correlation would be too easy: with 5,928 items, even r = .03 is "significant." So they tested agreement between individuals.

| Comparison (5,000 random pairs each) | Mean r | Meaning |
|---|---|---|
| Raters who saw the **same** 100 words | 0.136 (p = .04) | Real agreement |
| Raters who saw **different** words | 0.006 | No agreement (as expected) |
| Difference | d = 0.89 | Large |

**Example:** Riya and Sam rated the same 100 words, so their ratings should correlate if funniness is shared. Riya and Alex rated different words, so their correlation should be ~0.

## Check: is it just "everything is a 1"?

Repeated using only raters with ⩾ 25 ratings above 1: same-word r = **0.142**, different-word r = **0.002**, d = 0.94. Agreement isn't just shared boredom.

## The rude-word problem

- The funniest strings were rude-sounding: *whong*, *dongl*, *shart*, *focky*, *clunt* (plus *blablesoc*, 3rd funniest). This is **semantic parasitism**.
- Authors listed **52 rude substrings** (Appendix A: *ass, bum, poo, tit, dong, hump*…).
- **287 strings (4.92%)** contained one; **72%** of them were in the funnier half ($\chi^2 = 58.1$, $p < .0001$).
- This also shows people rated "funny ha-ha," not "funny peculiar." Rude-sounding strings are *less* peculiar (closer to real words), yet ranked funniest.

## Check: is the effect only rude words?

- After removing them: same-word r = **0.125** (p = .056, just short of significance), different-word r = **0.006**, d = 0.87.
- But **50.2%** of same-word pairs (2,510/5,000) were individually significant, vs. 2.5% expected by chance. Even using a realistic empirical chance rate of 12.5%, $p < .0001$.
- **Conclusion:** the effect is delicate but real. All later work uses the **5,641 clean strings**.

## Choosing the predictor

| Entropy computed over | Correlation with funniness |
|---|---|
| Single letters (unigrams) | 0.24 |

| Entropy computed over | Correlation with funniness |
|---|---|
| Letter pairs (bigrams) | 0.18 |
| Letter triples (trigrams) | 0.10 |

After controlling for unigrams, bigrams added 0.002% of variance and trigrams added nothing. **Unigram entropy** was chosen as the predictor.

## 6. Experiment stimuli and shared methods

The 200 test words were chosen only by their entropy number, never by human funniness ratings, so the experiments are true predictions.

### Selecting the stimuli

1. Generated **6,000 new** non-words, none from Study 1, all exactly **7 letters** (removes length effects).
2. Sorted by unigram entropy.
3. Took roughly every 60th string plus its nearest neighbor, skipping any that:
   - contained a rude substring,
   - had an ambiguous pronunciation,
   - broke English spelling rules.
4. Result: **200 strings in two matched sets of 100** (mean entropy 1.68, SD 0.19; correlation between sets 0.999). Listed in Appendix B.

| Predicted funny (low entropy) | Predicted unfunny (high entropy) |
|---|---|
| subvick (1.04) | tessina (2.02) |
| suppopp (1.11) | heashes (2.05) |
| quarban (1.34) | anceste (2.05) |
| himumma (1.34) | octeste (2.09) |

Notice the pattern: funny strings use rarer letters (*q, v, k, b, p, f, w, y*) and doubled consonants; unfunny ones are full of *e, s, t, a, n*.

### Participants

- **56** native English speakers (23 male, 33 female); mean age 19.9 (SD 3.3); ~13.6 years of education; course credit; ethics approved.
- Each did **both** experiments, using a different set of 100 words in each. Order of experiments and word sets was counterbalanced.

### Procedure

- Lab setting, ACTUATE software on Mac Minis, noise-reduced rooms.
- Instructions said the researchers had a model predicting humor and wanted to improve it.
- They used **"humorous"** instead of "funny" to avoid any "funny peculiar" reading.
- "This is not a test; we are just interested in your gut feeling."

### Analysis

- Generalized linear **mixed-effects models** in R; binomial models for choice data.
- **Random effects:** participant and stimulus order.
- **Predictors tried:** entropy, age, gender, education.
- Models compared with **AIC**.

## 7. Experiment 1: Forced choice

The bigger the entropy gap between two non-words, the more reliably people pick the lower-entropy one as funnier (from ~50% to ~85%).

### Task

Two strings side by side, lowercase, 60-pt Times. Press **x** if the left one is more humorous, **c** if the right one is. Answer quickly; if unsure, guess.

```
   subvick            octeste
  [press x]          [press c]
```

### Pairing design

- Highest entropy paired with lowest, second highest with second lowest, and so on.

- Produces pairs from **very different** (*subvick* 1.04 vs *octeste* 2.09, easy) to **nearly identical** (two words around 1.7, a coin flip).
- Each string appeared equally often on left and right.

## Predictions

1. The lower-entropy string is chosen as funnier.
2. This gets more reliable as the entropy gap grows.
3. Responses are faster for bigger gaps (easier decision).

## Data cleaning

- Removed responses < 400 ms (3.9%; too fast to be real judgments).
- Removed responses > 3 SD above mean (> 7,908 ms; 1.7%).
- One participant lost nearly half his data this way and was dropped → **55 participants**.

## Results: choices (Table 1)

| Model | AIC | Better than base? |
|---|---|---|
| Random effects only | 3579 | base |
| + age / education / gender | 3579–3581 | No |
| **+ entropy difference** | **3511** | **> 1,000,000×** |
| + random slopes | 3511–3512 | No |

- Demographics predict nothing; entropy difference predicts strongly.
- Fig. 3: chance of picking the predicted word rises roughly linearly from ~50% (no gap) to ~85% (largest gap).
- **22 of 55 (40%)** were individually reliable ($p < .05$); the best matched the prediction 76% of the time.
- **Top 50% of gaps:** 31 participants (56.4%) reliable; best = 92%.
- **Bottom 50% of gaps:** only 1 participant reliable, which is chance.

## Results: reaction times (Table 2)

- Bigger gaps → faster decisions ($t = 3.11$, $p = .002$), as predicted.

- Effect is **small**: AIC drops only 3 (2707 → 2704), so the entropy model is about 4.5× more likely than the base.

### Limitation

The pairing never put two low-entropy or two high-entropy words against each other. Experiment 2 fixes this by rating words one at a time.

## 8. Experiment 2: Rating task

Rated one at a time, lower-entropy strings get higher humor ratings across the whole range (model estimates vs. entropy: r = −0.83).

### Task

- Same 55 participants, rating the **100 strings they didn't see** in Experiment 1.
- One string at a time (40-pt Times) above an **unmarked slider**: "Least humorous" on the left, "Most humorous" on the right.
- Drag the marker; position recorded as **1 to 100**.

```
                quarban
Least humorous |---------------●-------| Most humorous
```

### Results (Table 3, Fig. 5)

| Model | AIC | Better than base? |
|---|---|---|
| Random effects only | 51,648 | base |
| + age / education / gender | 51,645–51,649 | No |
| **+ entropy** | **51,409** | **> 1,000,000×** |
| + random slopes | 51,412–51,413 | No |

- Lower entropy → higher rating (t = 15.6, p < .00001).
- Predicted rating falls from **~58/100** at entropy ~1.1 to **~32/100** at entropy ~2.1.
- Correlation between model estimates and entropy across all 200 strings: **−0.83**.
- Again, demographics predict nothing.

### What it adds

Confirms both main conclusions without the pairing limitation:

1. People agree on which non-words are funny.
2. How funny is predictable across the **entire** entropy range, not just extreme pairs.

## 9. General discussion

Non-words are funny because they violate our expectation of what a word is; entropy measures how far they violate it, and a humorist like Dr. Seuss lands on the same signature.

### Main contribution

First quantitative test of Schopenhauer's claim that humor scales with departure from an expected category, and the first statement of it in information-theory terms. Support from three angles: post-hoc fit (Study 1), forced choice (Exp. 1), individual ratings (Exp. 2).

### Theodor Lipps (1898) said it first

The authors found this idea independently, but Lipps had argued that funny pseudo-words work because we expect any word to mean something. The listener is "tricked for one moment," then realizes they were duped. This only works because prior language experience makes the form *look* meaningful.

### Dr. Seuss validation

- Seuss invented funny words on purpose: *wumbus*, *skritz*, *yuzz-a-ma-tuzz*.
- Per-letter entropy of **65 Seuss words** vs. **111,625 English words**: Seuss words are reliably lower ($t = 7.52$, $p < .00001$; Fig. 6).
- A humorist, working by intuition, picked exactly the property the model predicts (rare letters like *z, k, w, y*).

### Entropy is not the only cue

Other likely factors:

- Doubled letters (Seuss loves *zz, oo, kk*)
- Similarity to real words
- Unusual phonology (sound pattern)

- Other kinds of semantic parasitism beyond rude words

Humor is probably a synthesis of many cues.

## Why entropy and not raw letter frequency?

The two are closely correlated, so why the information-theory framing? Three reasons:

1. **Schopenhauer's theory is relational.** Humor depends on an event in the context of an expectation, not on a raw count. That is inherently information-theoretic (130 years before Shannon).
2. **Frequency is a shaky construct.** What should be counted? Baayen (2010) showed word-frequency effects can be fully explained by contextual measures. The mind probably doesn't keep raw counters; frequency effects emerge from **discriminative learning** (Rescorla–Wagner, 1972). Information-theoretic weighting also underlies:
   - **LSA:** log term frequency divided by the term's entropy across documents.
   - **tf–idf:** information in a term weighted by its occurrence probability (Aizawa, 2003).
   - **Gallistel (2003):** a cue is informative to the extent it reduces uncertainty.
3. **It generalizes.** In a pun or riddle there are no "letters" to count. Information theory gives one metric for any domain.

**Example extension:** the theory predicts "existential llama" (< 100 Google hits) is funnier than "angry llama" (> 13,000 hits).

## Why do violations feel funny at all?

- **Hurley, Dennett & Adams:** humor evolved to reward exploratory thinking ("bribing the brain with pleasure") and to signal it to others by laughing. Authors call their disagreement with Schopenhauer a "friendly amendment."
- Humor is likely a **multi-part meta-function**, not one brain process, which is why it is hard to pin down.
- **Brain evidence:**
  - Goel & Dolan (2001): puns and semantic jokes activate different regions; funniness tracks medial ventral prefrontal cortex.
  - Humor also activates the **dopamine reward system**: ventral striatum, nucleus accumbens, ventral tegmental area, amygdala.
  - Binder et al. (2006): the **fusiform gyrus** responds in steps to how word-like a string is, even with no task. It connects to the amygdala.

- **Speculative mechanism ("false alarm"):** the amygdala fires at an anomaly, then quickly switches off once it's judged harmless. Like realizing the figure in your backyard is just the neighbor's cat. Laughter signals to others: "I thought it was dangerous, but it's not."

### Link to sound symbolism

Non-word forms carry consistent intuitions in other ways too:

- **Köhler:** *maluma* = curvy, *takete* = spiky (ancestor of kiki–bouba).
- **Sapir (1929):** 80% linked *mal* (big vowel) to a large table, vs. *mil*.
- **Reilly et al. (2012):** people judge whether a non-word means something concrete or abstract.

### Possible applications

- Detecting changes in humor appreciation after brain damage or in mood disorders.
- **Product naming:** they predict *whook* or *mamessa* (funny) beat *turth* or *suppect* (unfunny).
- **Closing speculation:** real words also vary in entropy. People may prefer a low-entropy synonym for a light tone and a high-entropy one for a serious tone.

## 10. Limitations, critical notes and cheat sheet

The result is solid but narrow: entropy explains part of non-word humor, and the paper explains *that* violations are funny, not *why*.

### Limitations stated by the authors

- Entropy is one cue among many, even for non-words.
- They didn't ask participants *why* something was funny (people often can't say: "I dunno, it's just funny").
- Explains "why are non-words funny?" only abstractly; the deeper "why do violations feel funny?" stays open.
- Extending to real jokes is hard, because real-world co-occurrence probabilities are hard to estimate.
- The evolutionary and neural accounts are speculative and post-hoc.

## My critical observations

- **The effect is weak at the individual level.** Mean between-rater r ≈ 0.13; without rude words it is just short of p = .05. Strong at population level, delicate per person.
- **Mostly a letter-frequency effect.** Unigram entropy correlates strongly with simple letter frequency; the information-theory framing is argued on theoretical grounds more than shown to beat frequency empirically.
- **Terminology wobble.** Funny non-words are "low entropy," but the rare phrase "existential llama" is called "high entropy." Same idea (funnier = less probable), different label depending on summed component entropy vs. whole-item surprisal.
- **Length normalization unclear.** Text says divided by length; experiment values look summed. Harmless here (all 7 letters).
- **Sample:** young Canadian undergraduates; English only.
- **Rude-word filter is hand-made** (52 substrings the authors thought of), so some parasitism may remain.

## Key numbers cheat sheet

| What | Number |
|---|---|
| Study 1 participants / strings | 968 / 5,928 |
| Ratings per string | ~16 |
| Share of ratings that were "1" | 43.3% |
| Funniest / least funny | *whong* 5.92 / *exthe*, *suppect* 1.1 |
| Same-word vs. different-word rater r | 0.136 vs. 0.006 (d = 0.89) |
| Rude-alluding strings | 287 (4.92%); 72% in funnier half |
| Clean strings kept | 5,641 |
| Unigram / bigram / trigram correlation | 0.24 / 0.18 / 0.10 |
| Experiment participants | 56 (55 after cleaning) |
| Test stimuli | 200 (2 × 100), 7 letters, entropy 1.04–2.09 |
| Exp. 1 choice accuracy range | ~50% (no gap) to ~85% (max gap) |
| Exp. 1 reliable participants | 40% overall; 56.4% on top-half gaps |

| What | Number |
|---|---|
| Exp. 1 AIC gain (choice / RT) | > 1,000,000× / 4.5× |
| Exp. 2 rating range | ~58 to ~32 out of 100 |
| Exp. 2 model vs. entropy | r = −0.83 |
| Dr. Seuss words | 65, reliably lower entropy (t = 7.52) |

## One-line takeaway

Following Schopenhauer, humor = violated expectation; non-words violate "this is a word," and the rarer their letters (lower unigram entropy), the funnier people find them, predictably and regardless of age, gender or education.